

1. Network Layer Foundations & Switching

Mod 01

- **Role:** Host-to-Host end-to-end packet delivery across heterogeneous internetworks.
- **Store-and-Forward:** Router stores entire frame, verifies checksum, routes via table, and forwards.
- **Routing vs Forwarding:**
 - **Routing (Control Plane):** Software algorithms (OSPF, BGP) computing end-to-end optimal path.
 - **Forwarding (Data Plane):** Fast hardware engine transferring packet from input to output port.
- **Datagram vs Virtual Circuit (VC):**
 - *Datagram (Internet):* Connectionless, stateless routers, full 32-bit IP in every packet, dynamic reroute on failure, out-of-order delivery.
 - *Virtual Circuit (ATM/X.25):* 3-way handshake setup, stateful switches, 16-bit VCI tag, router crash kills circuit, guaranteed in-order.

3. IPv4 Fragmentation & MTU Mathematics

Mod 03

- **MTU (Maximum Transmission Unit):** Max L2 data payload (Ethernet MTU = 1500 Bytes).
- **Identification (16b):** Unique packet ID copied to all fragments for reassembly.
- **Flags (3b):**
 - Bit 0: Reserved (Must be 0).
 - Bit 1: **DF (Don't Fragment):** If 1 and packet > MTU, drop & send ICMP Type 3 Code 4.
 - Bit 2: **MF (More Fragments):** 1 = More fragments follow; 0 = Last fragment.

2. IPv4 Datagram Header Architecture (20-60B)

Mod 02

- **VER (4b):** Strictly 4 for IPv4 (0100 binary).
 - **HLEN (4b):** Header length in 4-byte words. Min = 5 (20B), Max = 15 (60B).
 - **ToS / DiffServ (8b):** Quality of Service priority bits.
 - **Total Length (16b):** Header + Payload in bytes. Max = 65,535B.
 - **TTL (8b):** Time-to-Live. Decrement by 1 at each router; at 0, packet dropped & ICMP Type 11 sent.
 - **Protocol (8b):** L4 multiplexing: 1=ICMP, 6=TCP, 17=UDP, 2=IGMP, 89=OSPF.
 - **Header Checksum (16b):** 1's complement sum of 16-bit words. Verifies header only; recomputed at every hop due to TTL.
- Payload Size = Total Length - (HLEN × 4) Bytes

4. IPv4 Classful Addressing Architecture & Special IP Blocks

Mod 04

Class	Leading Bits	1st Octet Range	NetID / HostID Bits	Default Mask	Networks / Hosts
A	0	1 – 126	8b / 24b	255.0.0.0 (/8)	126 / 16,777,214
B	10	128 – 191	16b / 16b	255.255.0.0 (/16)	16,384 / 65,534
C	110	192 – 223	24b / 8b	255.255.255.0 (/24)	2,097,152 / 254
D	1110	224 – 239	Multicast Group	N/A (No NetID/HostID)	Multicast addressing
E	1111	240 – 255	Reserved / Experimental	N/A	Research & Military

- **Private IPs (RFC 1918 - Non-routable):**
 - Class A: 10.0.0.0 – 10.255.255.255 (/8)
 - Class B: 172.16.0.0 – 172.31.255.255 (/12)
 - Class C: 192.168.0.0 – 192.168.255.255 (/16)
- **Special IP Addresses:**
 - 127.0.0.0/8 (127.0.0.1): Loopback testing.
 - 0.0.0.0: "This host on this network" (DHCP discover).
 - 255.255.255.255: Limited Broadcast (never routed).
 - NetID.All-1s: Directed Broadcast Address (DBA).

5. Subnetting Mechanics & Classful Design

Mod 05

- **Purpose:** Divides large wasteful host space into smaller departmental logical networks without changing external NetID.
- **Bit Borrowing:** Borrow s MSB bits from Host ID to form Subnet ID.
 - Subnets Formed = 2^s
 - Usable Hosts/Subnet = $2^h - 2$ (Subtract Subnet ID & DBA)
 - Block Size (Jump) = 256 - Interesting Mask Octet = 2^h
 - Router Subnet ID = Destination IP AND Subnet Mask
- **Example:** Class C 192.168.1.0 with 4 subnets $\implies s=2, h=6$. Mask = 255.255.255.192 (/26). Block size = 64. Subnets: .0, .64, .128, .192. Usable hosts = 62.

7. NAT & PAT (Port Address Translation)

Mod 07

- **NAT (RFC 1631):** Maps private RFC 1918 IPs to registered public IPs.
- **Types:**
 - *Static NAT:* 1:1 permanent mapping (Servers).
 - *Dynamic NAT:* Pool of public IPs assigned dynamically.
 - *PAT (NAT Overload):* Thousands of private IPs share **1 single public IP** using unique L4 Port numbers!
- **Translation Table:** Maps [Private IP:Port] $\rightarrow$ [Public IP:NAT Port].

9. Address Resolution Protocol (ARP) & RARP

Mod 08

- **Purpose:** Resolves Layer 3 Logical IP $\leftrightarrow$ Layer 2 Physical MAC.
- **ARP Request:** Broadcast (FF:FF:FF:FF:FF:FF): "Who has IP X? Tell Y".
- **ARP Reply:** Unicast back to requester: "I have IP X! My MAC is B".
- **ARP Cache:** RAM table storing (IP, MAC, TTL) to avoid repeated broadcasts. Dynamic entries age out (120-300s).

6. CIDR & Supernetting (Route Aggregation)

Mod 06

- **CIDR (RFC 1519):** Eliminates class boundaries. Notation IP/n.
- **3 Rules of CIDR Allocation:**
 1. Addresses in block must be contiguous.
 2. Total IPs $N = 2^{(32-n)}$ must be a power of 2.
 3. First address in decimal must be divisible by N .
- **Supernetting:** Aggregates contiguous Class C/CIDR routes into 1 entry (reduces global router tables). Prefix length decreases ($/24$ to $/22$).
- **Longest Prefix Match (LPM):** When multiple routes match a destination, router picks the one with the **longest prefix length** (most specific mask).

8. IPv6 Architecture & Transition

Mod 07

- **Address Space:** 128 bits (16 bytes) = 2^{128} approx 3.4 times 10^{38} IPs.
- **Base Header:** Fixed 40 Bytes. No checksum (removed), no broadcast (replaced by multicast/anycast).
- **8 Fields:** Version(4), Traffic Class(8), Flow Label(20), Payload Len(16), Next Hdr(8), Hop Limit(8), Src IP(128), Dst IP(128).
- **Transition Strategies:**
 - *Dual Stack:* Routers run both IPv4 and IPv6 stacks.
 - *Tunneling:* IPv6 encapsulated inside IPv4 packet.
 - *NAT-PT / Translation:* Header format translation gateways.

- **ARP 28B Packet:** HW Type (1=Eth), Prot Type (0x0800), HW Len (6), Prot Len (4), Opcode (1=Req, 2=Rep), Sender MAC/IP, Target MAC/IP.
- **Gratuitous ARP:** Host ARPs its own IP to detect IP conflicts or update neighbor caches on NIC change.
- **RARP:** MAC $\leftrightarrow$ IP resolution for diskless nodes (Replaced by DHCP).

10. Dynamic Host Configuration (DHCP)

Mod 09

- **Role:** Dynamic auto-assignment of IP, Mask, Gateway, DNS.
- **DORA Sequence (UDP Client:68, Server:67):**
 - **Discover:** Client broadcasts (255.255.255.255) from 0.0.0.0 .
 - **Offer:** Server proposes IP lease, mask, server ID.
 - **Request:** Client broadcasts accepting server's offer.
 - **ACK:** Server confirms lease with gateway and DNS info.
- **DHCP Relay Agent:** Router unicasts client broadcast across subnets.
- **Lease Timers:** T1 (50% time) unicast renew; T2 (87.5%) broadcast rebinding.

12. Distance Vector Routing & Count-to-Infinity

Mod 11

- **Algorithm:** Bellman-Ford: $D_x(y) = \min_v \{ c(x,v) + D_v(y) \}$.
- **RIP Protocol:** Periodic broadcast to immediate neighbors every 30s.
- **Count-to-Infinity Problem:** Link failure creates routing loops where nodes increment metric slowly to infinity (ping-pong bounce).
- **Mitigations:**
 - *Define Infinity* = 16 (Caps maximum loop count).
 - *Split Horizon:* Never advertise route back to neighbor learned from.
 - *Poison Reverse:* Advertise reverse route with metric = ∞ (16).

14. Path Vector (BGP) & Hierarchical Routing

Mod 13

- **BGP-4:** Inter-domain routing across Autonomous Systems (AS).
- **Path Vector:** Advertises complete AS-PATH: [AS200, AS100] .
- **Loop Discard Rule:** If router sees its own AS in AS-PATH, *discards update instantly!*
- **Hierarchical Routing:** Regions/clusters reduce tables from $O(N)$ to $O(\ln N)$.
Optimal Kamoun-Klein size: $S = 2 \cdot e \cdot \ln N$.

11. Internet Control Message Protocol (ICMP)

Mod 10

- **Role:** Network layer (Prot=1) error reporting and queries.
- **Error Types:**
 - *Type 3:* Destination Unreachable (Code 0=Net, 1=Host, 3=Port).
 - *Type 11:* Time Exceeded (Code 0=TTL Expired, 1=Reassembly).
 - *Type 12:* Parameter Problem. *Type 5:* Redirection.
- **Query Types:** Type 8 (Echo Req) & Type 0 (Echo Rep) $\implies$ ping .
- **Traceroute:** Sends packets with TTL=1, 2, 3... Intermediate routers drop packet and return ICMP Type 11. Target returns ICMP Type 3 Code 3.

13. Link State Routing & Dijkstra

Mod 12

- **Algorithm:** Dijkstra Shortest Path Tree (OSPF / IS-IS).
- **5 Phases:** (1) Discover neighbors via HELLO, (2) Measure link cost, (3) Build LSP, (4) Reliable Flooding of LSPs to all routers, (5) Run Dijkstra on complete LSDB.
- **DVR vs LSR:** DVR has local view and slow convergence (loops); LSR has global topology map, fast convergence, loop-free, but higher CPU/RAM overhead.

15. Congestion Control & Traffic Shaping

Mod 14

- **Open-Loop (Prevention):** Retransmission, window, drop policies.
- **Closed-Loop (Reaction):** Choke packets, backpressure, ECN bits.
- **Leaky Bucket:** Rigid constant output rate (Drops bursts).
- **Token Bucket:** Generates tokens at rate R \$, bucket capacity C \$. Allows bursts! Burst time: $T = C / (M - R)$.

16. Top 5 AKTU Semester Exam PYQs (10-Marks Long Questions)

Mod 14

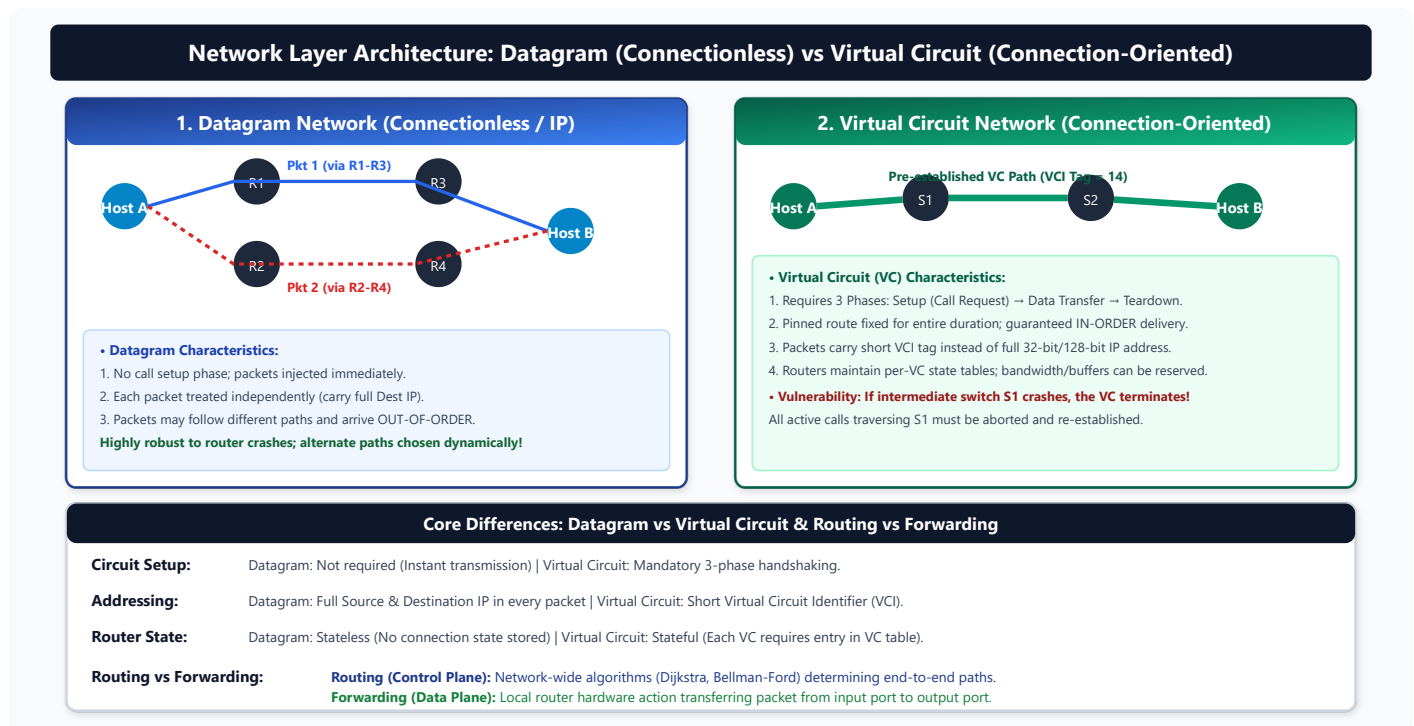
Question Topic	Key Derivation / Diagram to Draw	Ref Module
Datagram vs Virtual Circuit	Setup phase, stateless vs stateful, VCI tag, router failure rerouting table	Module 01
IPv4 Header Format & Fields	Draw 32-bit grid: VER, HLEN, Total Len, Identification, Flags, Offset, TTL, Prot, Checksum	Module 02
Subnetting Numericals	Formula $2^{\wedge} s \geq N$, borrow bits, jump size $= 256 - \text{ext}\{\text{mask}\}$, first/last usable IP, broadcast	Module 05
Count-to-Infinity & Solutions	Linear 3-node graph, link failure loop increment, Split Horizon with Poison Reverse	Module 11
Leaky vs Token Bucket	Draw water bucket with hole vs token bucket with tokens; burst formula $T = C / (M - R)$	Module 14

Module 01: Network Layer Foundations, Packet Switching & Virtual Circuits vs Datagram

Unit 3: Network Layer

Packet Switching & Switching Paradigms

End-to-end host delivery mechanics, Store-and-Forward packet switching, complete architectural comparison between Datagram and Virtual Circuit networks, and the Control Plane vs Data Plane separation.



1. Network Layer Responsibilities

The network layer provides end-to-end host-to-host packet delivery across heterogeneous internetworks. Key responsibilities include logical addressing (IP), path routing (control plane), local hardware forwarding (data plane), packetization, and fragmentation.

2. Datagram vs Virtual Circuit Switching

Architectural Philosophy

Datagram: Stateless routers, full IP header per packet, dynamic rerouting upon failure.

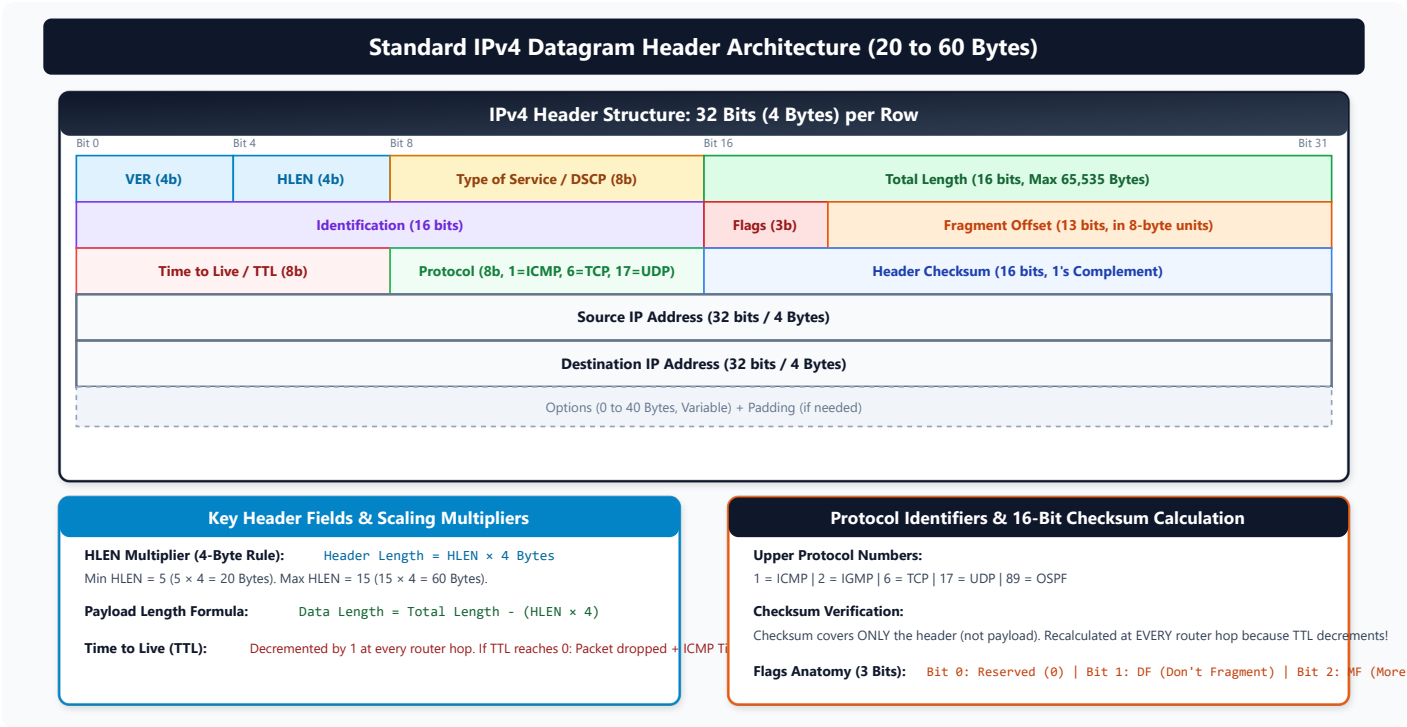
Virtual Circuit: Stateful switches, short VCI labels, pre-reserved bandwidth, single-router failure aborts circuit.

Module 02: IPv4 Header Format & Protocol Architecture

Unit 3: Network Layer

IPv4 Datagram Specification

Field-by-field architectural decomposition of the standard 20 to 60-byte IPv4 header, scaling factors, TTL loop protection, protocol multiplexing, and solved packet dissection numericals.



1. The 32-Bit Word Header Structure

IPv4 organizes header fields into 32-bit (4-byte) rows. HLEN measures length in 4-byte increments (5 to 15), dictating a header size between 20 and 60 bytes. Total Length spans up to 65,535 bytes.

2. Fragmentation and Lifetime Fields

Fundamental Equations

Header Size (Bytes) = $HLEN \times 4$

Payload Size = Total Length - $(HLEN \times 4)$

Byte Offset = Fragment Offset $\times 8$

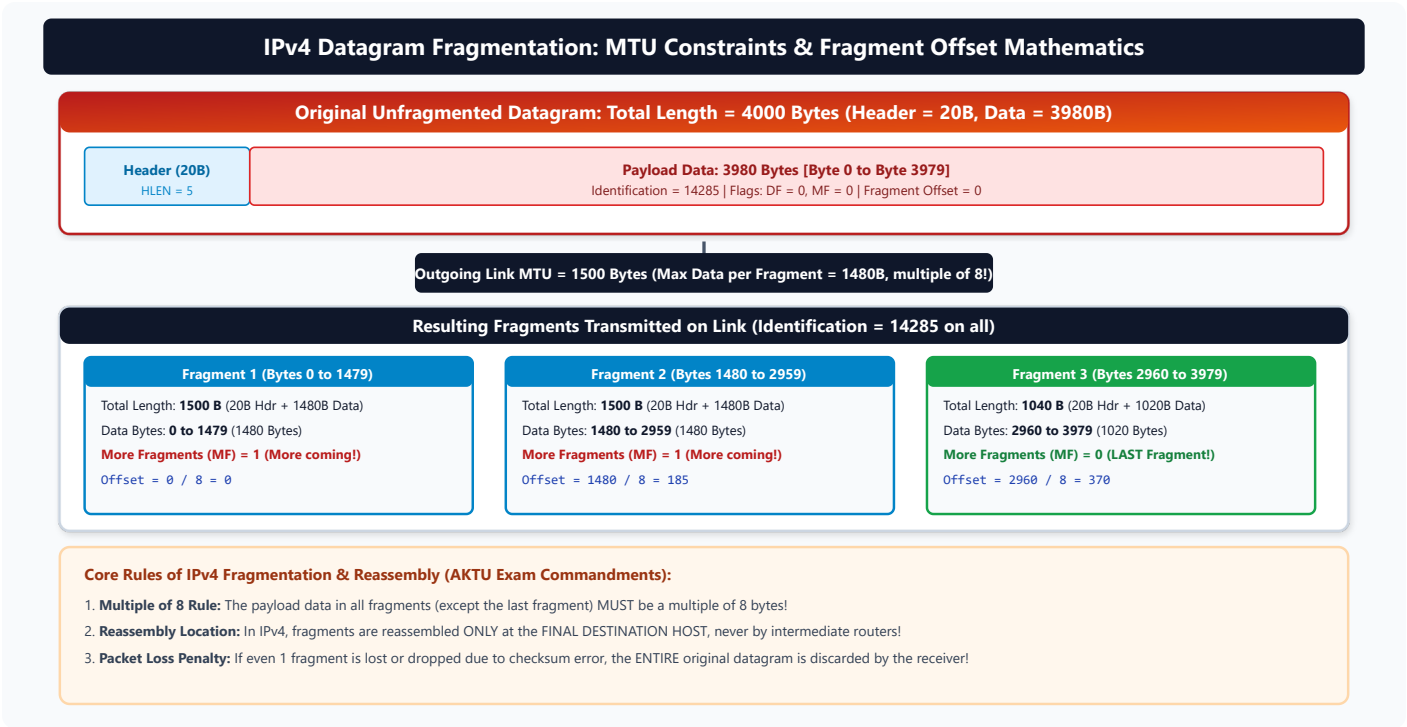
Hop Cost: $TTL_new = TTL_old - 1$ (Dropped when $TTL == 0$)

Module 03: IPv4 Fragmentation, Reassembly & Solved MTU Numericals

Unit 3: Network Layer

MTU & Offset Mathematics

Mathematical principles of IPv4 fragmentation across heterogeneous networks with differing MTUs, the 8-byte offset unit rule, destination-only reassembly, and end-to-end numerical derivations.



1. The MTU Boundary Constraint

When an IP datagram exceeds the Maximum Transmission Unit of an outgoing physical link, it is fragmented into smaller packets. Each fragment retains the original Identification number, an appropriate More Fragments (MF) flag, and a Fragment Offset in 8-byte increments.

2. Fragmentation Math Rules

Golden Fragmentation Laws

Max Data per Fragment = $\text{floor}((\text{MTU} - \text{Header}) / 8) \times 8$

Fragment Offset = Starting Byte Index / 8

MF = 1 for all fragments except the last fragment (MF = 0)

Module 04: IPv4 Classful Addressing & Special Purpose Blocks

Complete mathematical exploration of IPv4 Class A, B, C, D, and E architectures, NetID vs HostID divisions, default subnet masks, usable host calculations ($2^H - 2$), and RFC 1918 private address ranges.

IPv4 Classful Addressing Hierarchy: Classes A, B, C, D & E Breakdown

1. Class Bit Patterns & Address Ranges

0	NetID (7b)	HostID (24 bits = 16,777,214 Hosts)
10	NetID (14 bits = 16,384 Nets)	HostID (16 bits = 65,534 Hosts)
110	NetID (21 bits = 2,097,152 Nets)	Host (8b: 254)
1110	Multicast Group Address (28 bits)	
1111	Reserved for Future / Research & Development	

Default Subnet Masks:
Class A: 255.0.0.0 (/8) | Class B: 255.255.0.0 (/16) | Class C: 255.255.255.0 (/24)

2. Class Decimal Ranges & Special Blocks

Class A	0.0.0.0 - 127.255.255.255	Leading bits: 0...
Class B	128.0.0.0 - 191.255.255.255	Leading bits: 10...
Class C	192.0.0.0 - 223.255.255.255	Leading bits: 110...
Class D	224.0.0.0 - 239.255.255.255	Multicast Groups
Class E	240.0.0.0 - 255.255.255.255	Experimental / R&D

RFC 1918 Private IPv4 Ranges (Non-Routable):

- Class A: 10.0.0.0 to 10.255.255.255 (10.0.0.0/8)
- Class B: 172.16.0.0 to 172.31.255.255 (172.16.0.0/12)
- Class C: 192.168.0.0 to 192.168.255.255 (192.168.0.0/16)

Conserves global IPv4 space; requires NAT for internet access!

Special Purpose IPv4 Addresses (Never Assigned to Regular Host NICs)

This Host (0.0.0.0):

Used by host during bootstrapping (DHCP Discover) when it doesn't yet know its own IP address.

Loopback (127.0.0.1 - 127.x.x.x):

Packets never leave the local machine. Used to test local TCP/IP stack software (localhost).

Limited Broadcast (255.255.255.255):

Broadcast to all hosts on the local network segment. Strictly NEVER forwarded by routers!

Directed Broadcast (HostID = All 1s):

E.g., 192.168.1.255. Delivered to all hosts in that specific remote network.

Network ID (HostID = All 0s):

E.g., 192.168.1.0. Identifies the entire network itself; cannot be assigned to an individual host interface.

1. The 5 Address Classes

Classful addressing divides the 32-bit address space based on leading bits: Class A (0, /8), Class B (10, /16), Class C (110, /24), Class D (1110, Multicast), and Class E (1111, Experimental).

2. Usable Host Capacity Equation

Address Exclusion Rules

Usable Hosts = $2^H - 2$

Excluded: Network Address (HostID = All 0s) and Directed Broadcast (HostID = All 1s)

Private IP Blocks: 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16

Module 05: Subnetting Mechanics and Classful Subnet Design

Unit 3: Network Layer

Subnetting & Bit Borrowing

Concept of Subnetting, Host-Bit Borrowing, Subnet Mask Math, Block Size Formula, Step-by-Step Numerical Calculations & Bitwise AND Subnet ID Extraction.

Subnetting Architecture: Bit Borrowing & Subnet ID Formation

1. Original Class C IP Structure (/24 Mask: 255.255.255.0)

Network ID (24 Bits) - Fixed by ISP / IANA

Host ID (8 Bits) = 256 IPs

Borrow s = 2 Bits from Host ID

2. Subnetted Structure (/26 Mask: 255.255.255.192)

Network ID (24 Bits)

Subnet (2 Bits)

Host ID (6 Bits)
2^6 - 2 = 62 Hosts

Bitwise AND: Router Subnet Extraction

IP: 192.168.1.75 11000000.10101000.00000001.01001011
Mask: 255.255.255.192 11111111.11111111.11111111.11000000
AND: 192.168.1.64 11000000.10101000.00000001.01000000
Result: Packet belongs to Subnet 1 (Range: .64 to .127)

4 Subnets Generated (Jump Size = 64)

Subnet ID	First Usable IP	Last Usable IP	Broadcast ID
.0 (Sub 0)	192.168.1.1	192.168.1.62	192.168.1.63
.64 (Sub 1)	192.168.1.65	192.168.1.126	192.168.1.127
.128 (Sub 2)	192.168.1.129	192.168.1.190	192.168.1.191
.192 (Sub 3)	192.168.1.193	192.168.1.254	192.168.1.255

1. Fundamental Formulas

Subnet Engineering Rules

Number of Subnets = 2^s (where s = borrowed bits)

Valid Hosts per Subnet = 2^h - 2 (where h = remaining host bits)

Block Size (Jump Size) = 256 - Mask Octet Value = 2^h

Subnet ID = Destination IP Bitwise AND Subnet Mask

2. Subnet Allocation Table (Class C /26 Example: 193.1.2.0)

Subnet	Subnet ID	First Usable IP	Last Usable IP	Directed Broadcast
Subnet 0	193.1.2.0	193.1.2.1	193.1.2.62	193.1.2.63
Subnet 1	193.1.2.64	193.1.2.65	193.1.2.126	193.1.2.127
Subnet 2	193.1.2.128	193.1.2.129	193.1.2.190	193.1.2.191
Subnet 3	193.1.2.192	193.1.2.193	193.1.2.254	193.1.2.255

Module 06: Classless Addressing (CIDR) and Supernetting

Classless Inter-Domain Routing (CIDR), 3 Mandatory Block Allocation Rules, Supernetting Mechanics, and Longest Prefix Match (LPM) Routing Forwarding Algorithm.

CIDR Slash Notation, Block Rules & Supernetting (Route Aggregation)

1. CIDR Address Breakdown: 200.16.0.0 / 22

Network Prefix: n = 22 Bits (Mask: 255.255.252.0)

Host Bits: 32 - n = 10 Bits (1024 IPs)

3 Mandatory Rules for CIDR Allocation

- Rule 1 (Contiguous): All addresses must be continuous.
- Rule 2 (Power of 2): Total IPs $N = 2^{(32 - n)}$ (e.g. 1024, 2048).
- Rule 3 (Divisible): First IP (in decimal) must be divisible by N.

Supernetting: 4 Class C Networks Aggregated

198.16.0.0/24

198.16.1.0/24

198.16.2.0/24

198.16.3.0/24

↓ Merged into Single Routing Table Entry ↓

Supernet: 198.16.0.0 / 22 (Mask: 255.255.252.0)

Longest Prefix Match (LPM) Routing Table Decision Rule

Destination Network	Prefix Length	Next Hop / Interface	Packet: 200.24.16.99 (Evaluation)
200.24.0.0	/16 (255.255.0.0)	Interface eth0	Matches (16-bit match)
200.24.16.0	/20 (255.255.240.0)	Interface eth1	Matches (20-bit match)
200.24.16.64	/26 (255.255.255.192)	Interface eth2	Selected! Longest Prefix (26 bits)

1. 3 Mandatory Rules of CIDR Allocation

- Rule 1: Addresses in a block must be contiguous.
- Rule 2: Total count of addresses must be power of 2 ($N = 2^{(32 - n)}$).
- Rule 3: First address (in decimal integer) must be divisible by block size N.

2. Subnetting vs Supernetting Comparison

Parameter	Subnetting	Supernetting
Objective	Divide large network into smaller subnets	Combine smaller contiguous networks
Prefix Mask	Prefix increases (/24 to /26)	Prefix decreases (/24 to /22)
Routing Table	Internal routing precision	Compacts global Internet routing table

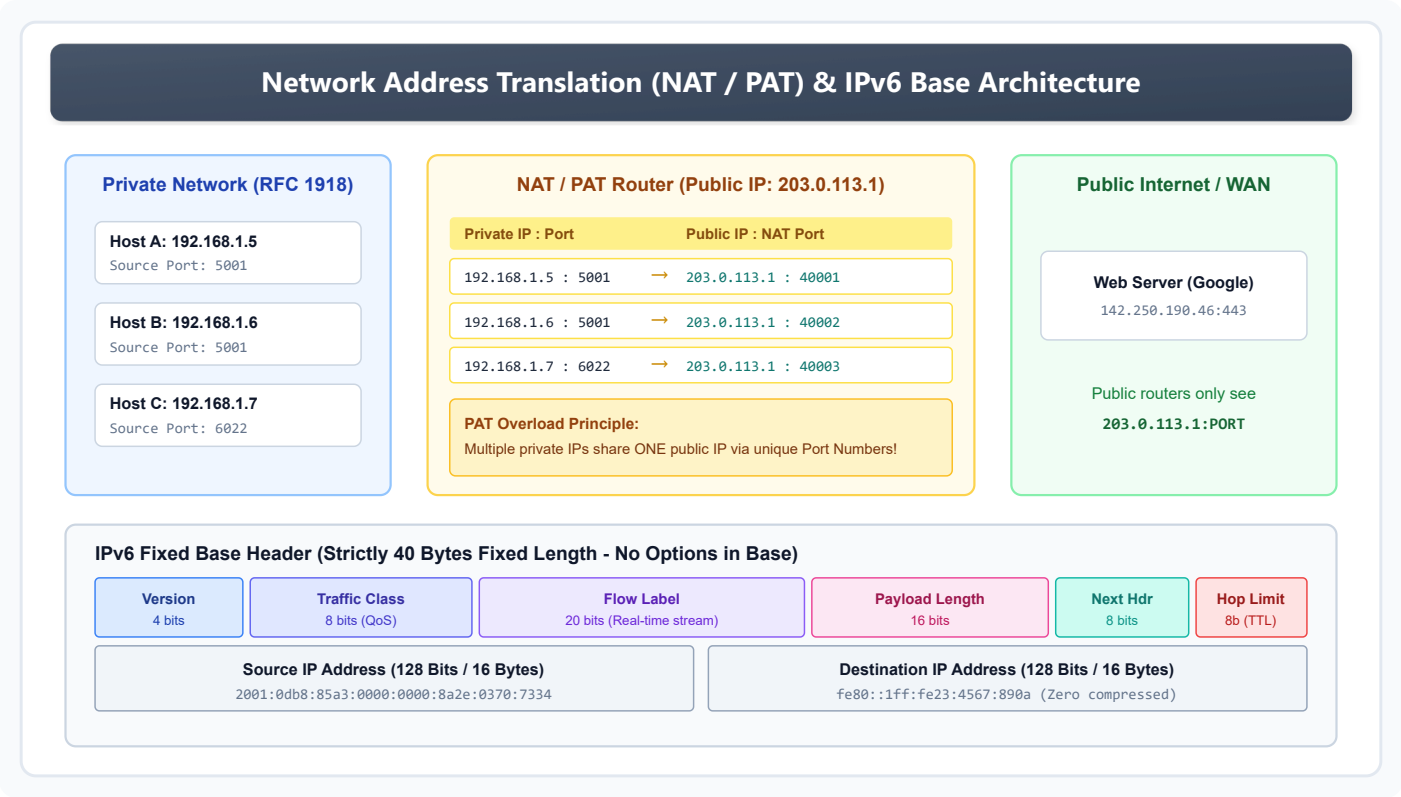
3. Longest Prefix Match (LPM)

When multiple routing prefixes match an incoming destination IP address, the router selects the entry with the **highest prefix length (most specific mask)**.

Module 07: Network Address Translation (NAT), PAT, and IPv6 Architecture

Unit 3: Network Layer NAT/PAT & IPv6 Evolution

Private-to-Public IP Translation, Port Address Translation (PAT) Overload, IPv6 128-bit Addressing Architecture, 40-Byte Fixed Base Header, and Transition Strategies.



1. NAT Translation Types

- Static NAT: 1-to-1 permanent private-to-public mapping.
- Dynamic NAT: Pool of public IPs assigned on first-come first-served basis.
- PAT (Port Address Translation / NAT Overload): Thousands of private hosts share 1 single public IP using unique L4 Port numbers!

2. IPv4 vs IPv6 Architectural Comparison

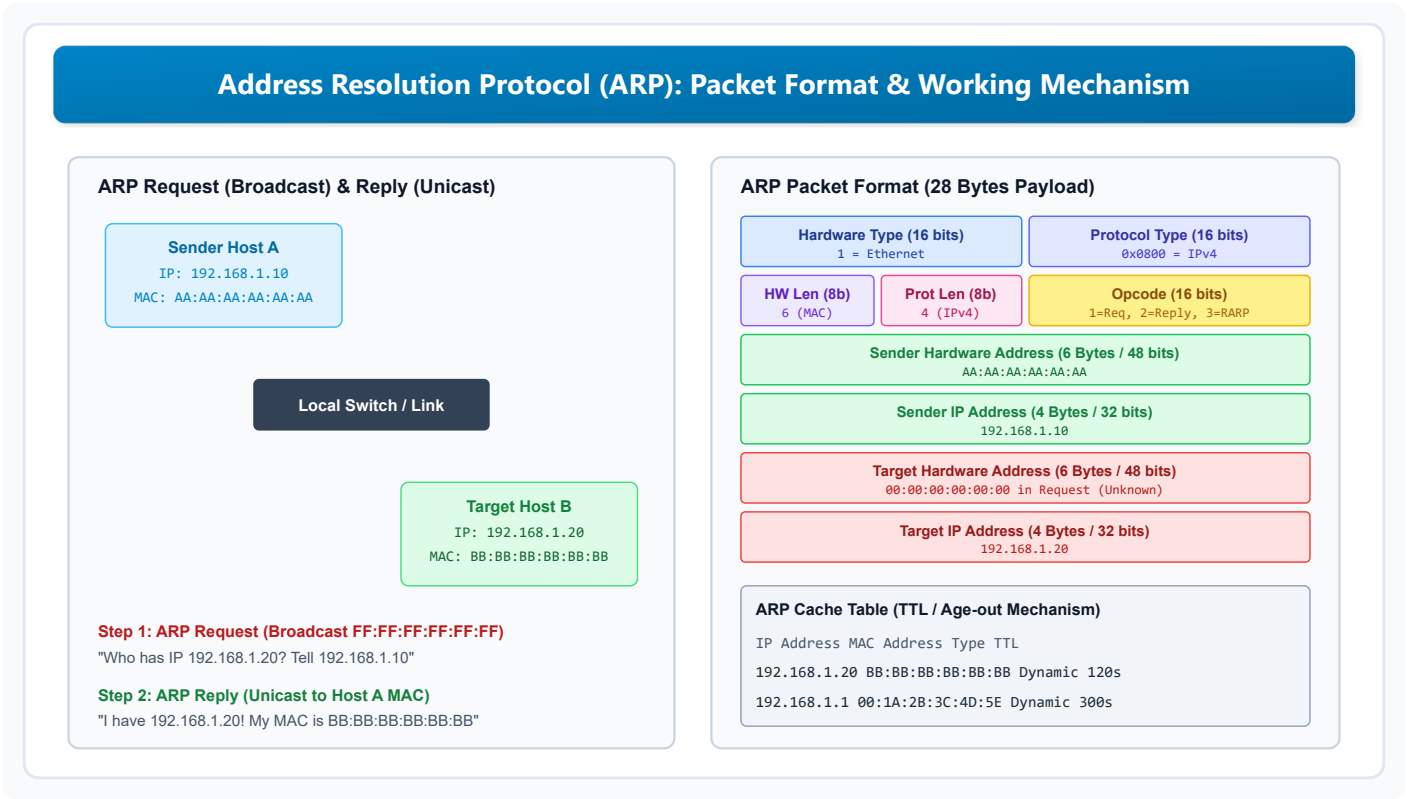
Feature	IPv4	IPv6
Address Length	32 bits (4 bytes)	128 bits (16 bytes)
Header Size	Variable: 20 to 60 Bytes	Fixed: Strictly 40 Bytes
Header Checksum	Present (recomputed at every hop)	Eliminated (handled by L2 & L4)
Fragmentation	Routers & Source host	Source host only (Path MTU Discovery)
Broadcast	Supported (Limited & Directed)	Eliminated (Replaced by Multicast/Anycast)

Module 08: Address Resolution Protocol (ARP) and RARP

Unit 3: Network Layer

L3 to L2 Address Resolution

Logical IP to Physical MAC Address Binding, ARP Request Broadcast & Reply Unicast, 28-Byte Packet Format, 4 Operational Scenarios, ARP Cache, Gratuitous ARP & RARP.



2. ARP 28-Byte Packet Anatomy

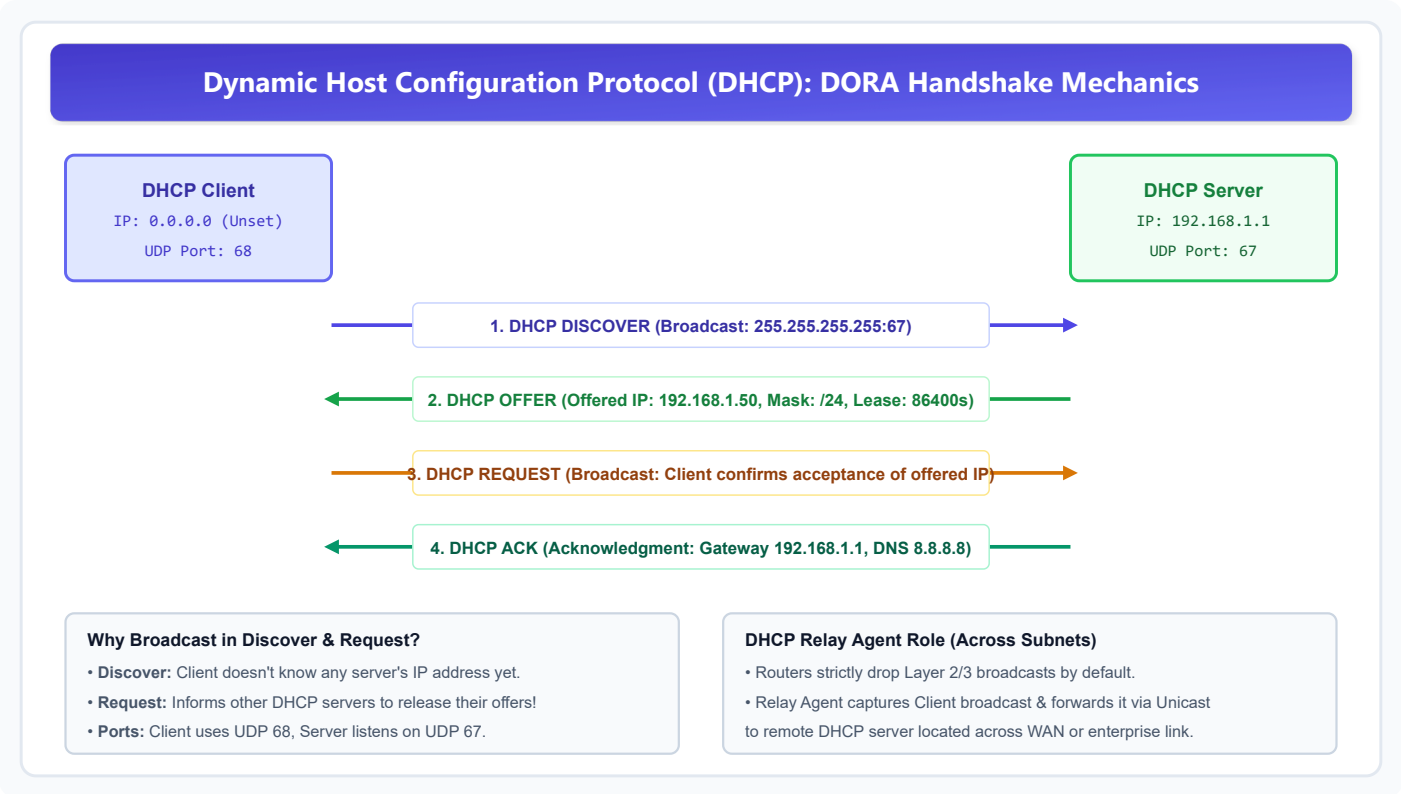
Field	Bit Width	Description / Sample Value
Hardware Type	16 bits	1 for Ethernet link
Protocol Type	16 bits	0x0800 for IPv4 protocol
HW / Prot Length	8 + 8 bits	6 bytes (MAC) / 4 bytes (IPv4)
Opcode	16 bits	1 = Request, 2 = Reply, 3 = RARP Req, 4 = RARP Rep
Sender HW / IP Addr	48 + 32 bits	Sender's MAC (6B) and IP (4B)
Target HW / IP Addr	48 + 32 bits	Target MAC (00:00:... in Request) and Target IP (4B)

Module 09: Dynamic Host Configuration Protocol (DHCP)

Unit 3: Network Layer

Automatic Host Configuration

Dynamic IP Allocation Architecture, DORA 4-Step Handshake, UDP Port 67/68 Design, Lease Renewal Timers, and DHCP Relay Agent Mechanics.



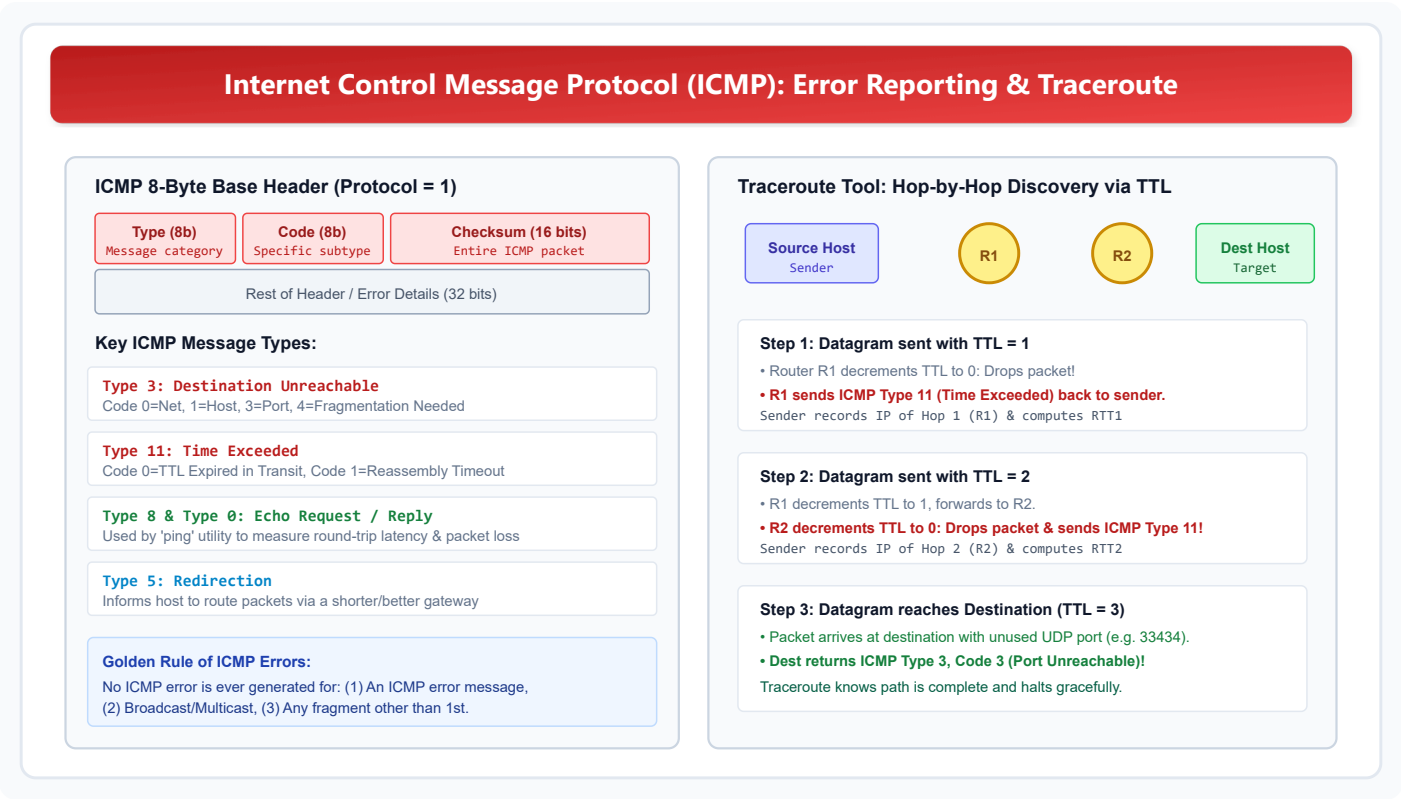
O	DHCP Offer	Server -> Client	Src Server_IP:67 -> Dst Client/Broadcast:68	Offers candidate IP, mask, lease duration
R	DHCP Request	Client -> Server	Src 0.0.0.0:68 -> Dst 255.255.255.255:67	Broadcast accepting one server's offer
A	DHCP ACK	Server -> Client	Src Server_IP:67 -> Dst Client/Broadcast:68	Finalizes lease with gateway and DNS info

Module 10: Internet Control Message Protocol (ICMP)

Unit 3: Network Layer

Network Diagnostics & Error Control

Architectural role of ICMP, Error Reporting vs Query Types, 8-Byte Header Layout, Ping Diagnostics, and Traceroute TTL Discovery Mechanics.



5	0 - 3	Redirection	Error	Route optimization update
8	0	Echo Request	Query	Ping query transmitted by sender
11	0	TTL Expired in Transit	Error	TTL reached zero (Traceroute hop discovery)

Module 11: Distance Vector Routing and Count-to-Infinity

Unit 3: Network Layer

Bellman-Ford & Routing Dynamics

Mathematical Foundation of Distance Vector Routing, Bellman-Ford Recurrence, The Count-to-Infinity Failure Phenomenon, and Split Horizon / Poison Reverse Mitigations.

Distance Vector Routing (Bellman-Ford) & Count-to-Infinity Problem

$$\text{Bellman-Ford Equation: } D_x(y) = \min_v \{ c(x,v) + D_v(y) \}$$

Linear 3-Node Topology (RIP Protocol)



Stable Routing Vectors (to Destination A):

- At Node B: Distance to A = 1 (Direct via link B-A)
- At Node C: Distance to A = 2 (Next Hop = B)

Sudden Link Failure: Link A-B Goes DOWN!

- B detects link A-B is dead.
- But before B can advertise this, C tells B: "I have route to A (cost=2)!"
- B mistakenly thinks: "Great! I can reach A via C with cost = 2+1 = 3!"
- Routing Loop created: B points to C, C points to B!

Count-to-Infinity Ping-Pong Cycle

Exchange Step	B's Cost to A	C's Cost to A
Iteration 1	3 (via C)	2 (via B)
Iteration 2	3 (via C)	4 (via B)
Iteration 3	5 (via C)	4 (via B)
...	Continues until metric reaches 16 (Infinity)	

Standard Engineering Solutions:

- Defining Infinity (RIP = 16):**
Caps maximum loop counting at 16 hops.
- Split Horizon:**
Never advertise a route back to the neighbor from whom it was learned!
- Split Horizon with Poison Reverse:**
Explicitly advertise the reverse route with cost = infinity (16).

1. Bellman-Ford Recurrence Relation

$$D_x(y) = \min_v \{ c(x,v) + D_v(y) \}$$

Periodic Updates: Broadcast entire routing table to direct neighbors every 30 seconds.

Metric Cap: RIP defines Infinity = 16 hops to limit count-to-infinity duration.

2. Count-to-Infinity Mitigations

Technique	Core Mechanism	Tradeoff
Hop Count Cap (16)	Caps maximum routing loop iterations	Limits network diameter to 15 hops
Split Horizon	Never advertise a route back to the advertising neighbor	Prevents 2-node loops
Poison Reverse	Explicitly advertises reverse route with metric = Infinity (16)	Instant loop elimination

Module 12: Link State Routing and Dijkstra Algorithm

Link State Protocol Architecture (OSPF/IS-IS), 5 Execution Phases, LSP Reliable Flooding, Step-by-Step Dijkstra Numerical, and Master DVR vs LSR Comparison.

Link State Routing (LSR): LSP Reliable Flooding & Dijkstra's Algorithm

5 Phases of Link State Routing (OSPF / IS-IS)

1. Discover Neighbors:
Send HELLO packets to learn local interface neighbours.

2. Measure Link Metrics:
Send ECHO packets to measure round-trip delay, bandwidth.

3. Build Link State Packet (LSP):
Create LSP containing Node ID, sequence no, & neighbor costs.

4. Reliable Flooding of LSPs:
Distribute LSP to every router in the AS using ACKs.

5. Compute Shortest Path Tree:
Every node runs Dijkstra algorithm on its complete LSDB.

Structure of a Link State Packet (LSP):

- Source Router ID: R1
- Sequence Number: 0x0000004A (Prevents old packet reuse)
- Age: 3600 seconds (Decrement to prevent zombie entries)
- Neighbor 1: R2 (Metric = 2)
- Neighbor 2: R3 (Metric = 5)
- Neighbor 3: R4 (Metric = 1)

Dijkstra Shortest Path Tree (Source = Node U)

```
graph LR; U((U)) ---|2| V((V)); U ---|5| W((W)); V ---|1| X((X)); W ---|2| X
```

Step-by-Step Dijkstra Trace:

Step	Confirmed Set N'	D(V), p(V)	D(W), p(W)	D(X), p(X)
0	{U}	2, U*	5, U	inf
1	{U, V}	-	5, U	3, V* (2+1)
2	{U, V, X}	-	5, U*	-

Resulting Routing Table at Router U:

- Dest V: Cost = 2, Next Hop = V (Direct link)
- Dest X: Cost = 3, Next Hop = V | Dest W: Cost = 5, Next Hop = W

1. 5 Phases of Link State Routing

1. Discover neighbors via periodic HELLO packets.

2. Measure link cost/latency via ECHO test.

3. Construct Link State Packet (LSP) with Seq Number & Age.

4. Reliable Flooding of LSP to all routers in Autonomous System.

5. Execute Dijkstra's Algorithm independently on Link State Database (LSDB).

2. DVR vs LSR Master Comparison

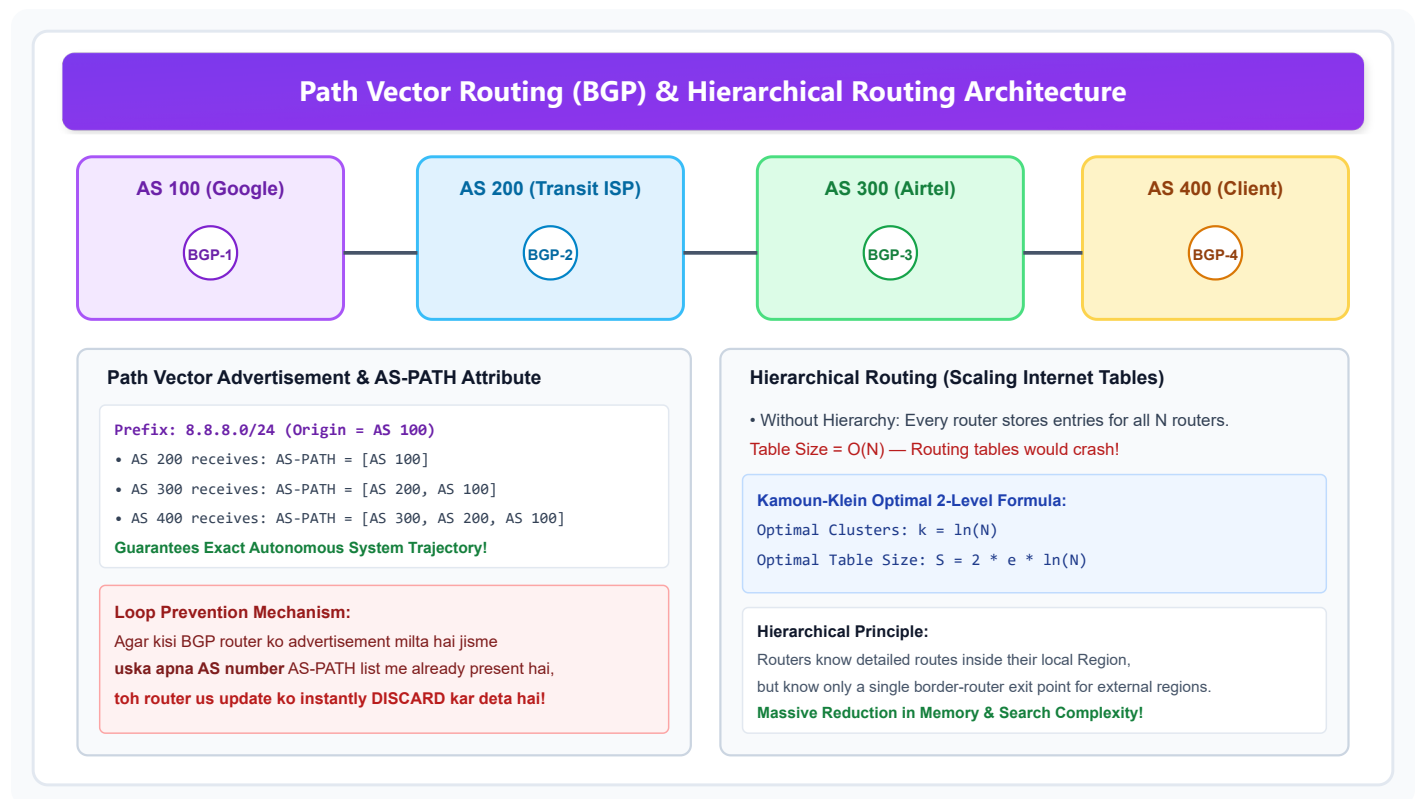
Feature	Distance Vector (RIP)	Link State (OSPF)
Graph Knowledge	Local view only (Neighbor vectors)	Complete global topology (LSDB)
Update Sharing	Shares entire table with neighbors	Shares neighbor state with entire network
Convergence	Slow (Count-to-Infinity loops)	Fast & mathematically loop-free
Resource Overhead	Minimal memory & CPU	High CPU for Dijkstra graph execution

Module 13: Path Vector, Hierarchical Routing, and BGP

Unit 3: Network Layer

Inter-Domain & Hierarchical Scaling

Inter-Domain Routing Foundations, Border Gateway Protocol (BGP-4), Path Vector AS-PATH Attribute, Loop Prevention, and Kamoun-Klein Hierarchical Clustering.



1. BGP Path Vector & Loop Prevention

Path Vector Advertisement: Carries complete sequence of Autonomous Systems (AS-PATH).

Loop Prevention Rule: If receiving BGP speaker detects its own AS in AS-PATH, discard route immediately!

Policy Routing: Routes chosen by commercial contracts & security policies rather than raw latency.

2. Hierarchical Routing Scaling

Routing Model	Table Entries	Update Frequency	Scalability Limit
Flat Routing	$O(N)$ (One per router)	Floods entire graph	Crashes at > 10,000 routers
Hierarchical (2-Level)	$O(\ln N)$ (Optimal Kamoun-Klein)	Localized to region	Scales to millions of Internet nodes

Module 14: Congestion Control, QoS, and Unit 3 AKTU PYQs

Congestion Control Paradigms, Open-Loop vs Closed-Loop Techniques, Leaky Bucket vs Token Bucket Traffic Shaping, and Solved AKTU Semester Exam PYQs.

Network Layer Congestion Control: Traffic Shaping & Bucket Algorithms

1. Leaky Bucket Algorithm (Traffic Smoothing)

Core Working Principle:

- Input can arrive in bursts of any rate.
- If buffer fills completely, extra packets are **DISCARDED**.
- Output flow is strictly constant (Cannot send bursts!).

2. Token Bucket Algorithm (Allows Controlled Bursts)

- Tokens generated at fixed rate **R** tokens/sec.
- Bucket accumulates up to maximum Capacity **C** tokens.
- To transmit **K** bytes, host must grab **K** tokens.

Burst Duration Formula (AKTU Numerical):
Burst Time: $T = C / (M - R)$
(**M** = Peak Transmission Rate, **R** = Token Generation Rate)

Closed-Loop Congestion Control Mechanisms:

1. Choke Packets:
Congested router generates ICMP packet directly to sender to slow down.

2. Hop-by-Hop Backpressure:
Upstream router throttles previous neighbor link immediately.

3. Explicit Congestion Notification (ECN):
Router marks ECN bits in IPv4/IPv6 header (RFC 3168).
Prevents packet drops by proactively informing TCP endpoints!

1. Leaky Bucket vs Token Bucket Comparison

Feature	Leaky Bucket	Token Bucket
Output Stream	Strictly constant rate (Smooth)	Allows controlled bursty output
Discard Behavior	Discards packets on full buffer	Discards tokens on full bucket (Zero packet loss)

Burst Transmission	Cannot transmit bursts	Burst time: $T = C / (M - R)$
---------------------------	------------------------	-------------------------------

2. Closed-Loop Feedback Mechanisms

1. Choke Packets: Router generates direct ICMP warning message to sending host.
2. Hop-by-Hop Backpressure: Immediate upstream link throttling.
3. Explicit Congestion Notification (ECN): Marks IP header bits without packet dropping.